Development of a machine learning model to classify different postures for the sitting position Portuguese Conference on Pattern Recognition

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Abstract

This paper presents the development of a Machine Learning model aimed at classifying sitting postures, with the broader goal of promoting ergonomic practices and preventing musculoskeletal disorders in the workplace. The motivation behind this study lies in the growing concern about health problems related to prolonged incorrect postures during desk work. Data were collected using pressure sensors embedded in an ergonomic chair, capturing ten distinct sitting postures, each sustained for 30 seconds. Data analysis was performed on the Google Colab platform with the PyCaret library, which enabled efficient testing and comparison of multiple classification algorithms. The preliminary results indicate a high potential for accurate posture detection. A comparative evaluation of model performance and sensor data relevance is ongoing. The findings highlight the feasibility of integrating intelligent posture monitoring systems into everyday office environments, contributing to the advancement of ergonomics at work and proactive health monitoring.

1 Introduction

Musculoskeletal disorders (MSDs) continue to pose a major challenge in occupational health, particularly in environments that require extended periods of sitting, such as control rooms, office spaces, and call centers. These conditions, often associated with sustained poor postural habits, not only impair individual well-being, but also reduce productivity and increase absenteeism [1][2].

2 State of the art

The increasing incidence of MSDs, particularly in occupations requiring prolonged static postures, has driven interest in intelligent systems capable of tracking and improving seated posture. Traditional ergonomic assessments, which rely on observational checklists or self-reports, are often subjective and lack real-time responsiveness, reducing their effectiveness in dynamic workplace settings.

2.1 Posture Monitoring with Pressure Sensors and Machine Learning

Pressure sensors enable non-intrusive and continuous posture monitoring by capturing pressure distribution across seating surfaces, allowing accurate detection and classification of user positions. Recent advances combine these sensors with machine learning algorithms—such as KNN, SVM, and CNN—to automatically recognize sitting postures with high accuracy. For example, Tsai et al. (2023) achieved over 99% accuracy using ensemble learning, while Lee and Kim (2022) applied CNNs to classify nine children's postures with high precision. These systems are increasingly applied in ergonomic and health-related contexts, providing real-time feedback and promoting correct sitting habits. Despite promising results, challenges persist regarding user-specific calibration, system generalization, and seamless integration into everyday environments.

3 Metodology

This section outlines the experimental procedures, data acquisition strategy, and modeling approach used in this study. A systematic framework was adopted to ensure consistent data collection and evaluate the performance of machine learning algorithms in classifying seated postures based on input from the pressure sensor.

Models	No Filter	Filter	No Filter + BMI	Filter + BMI
RF	0.9355	0.9343	0.9475	0.9391
XGB	0.9152	0.9140	0.9677	0.9761
SVM	0.9390	0.8866	0.2186	0.1637
ANN	0.9463	0.9379	0.9486	0.9427
KNN	0.5247	0.4780	0.7193	0.6727

Table 1: Models Accuracy

3.1 Data Acquisition

Following approval from the Ethics Committee of Coimbra Polytechnic University, a total of 40 volunteers participated in the study, helping collect data using an instrumented chair equipped with 32x32 pressure sensors distributed throughout the seating surface (Fig. 1).

As it can be seen in Fig. 2, each volunteer was instructed to simulate 10 distinct sitting postures. Each posture was held for 30 seconds. The selected postures focus on the pressure-based ergonomic assessment for the seated configurations.

3.2 Data Processing and Model Evaluation

A median filter was applied to pressure map images to reduce noise while preserving structural details. Additionally, three features—Body Mass Index (BMI), Center of Pressure (CoP), and Percentage of Pressure—were extracted and tested individually or combined with raw and filtered images to assess their influence on classification accuracy. The classification workflow was implemented in PyCaret using Google Colab. Five algorithms—Random Forest, XGBoost, SVM, ANN, and KNN—were trained with different input configurations and validated through 10-fold cross-validation. Model performance was evaluated using accuracy and F1-score to ensure robust, balanced multiclass posture classification.

4 Results

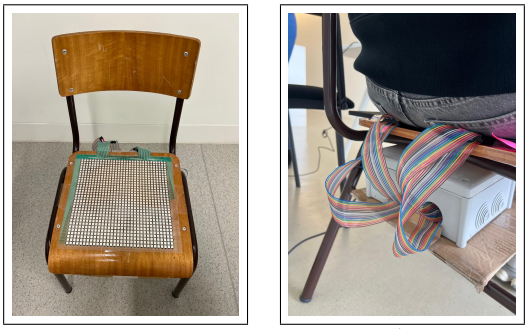
This section presents the performance of various machine learning models applied to posture classification using pressure map images and derived features.

4.1 Performance with Raw Data and BMI

4.1.1 Accuracy

The initial experiments were conducted using raw pressure map data across the five models. In the Table 1, the ANN model achieved the highest baseline accuracy, with SVM and RF also showing strong initial results.

To evaluate the impact of the physiological context, BMI was introduced as an external feature. Its inclusion led to a significant improvement in accuracy, especially in tree-based models. XGB, in particular, achieved the highest accuracy when used with filtered images and BMI. RF also benefited noticeably. SVM, however, suffered a steep performance drop, likely due to feature scale sensitivity and the non-linear effect of BMI.



(a) (b)

Figure 1: Data acquisition setup: (a) Chair equipped with a matrix of 32x32 pressure sensors; (b) Participant during a data collection procedure.



(a)

Figure 2: Postures used in data collection [3].

Models	No Filter	Filter	No Filter + BMI	Filter + BMI
RF	0.9305	0.9282	0.9430	0.9332
XGB	0.9085	0.9079	0.9651	0.9750
SVM	0.9343	0.8781	0.1580	0.1149
ANN	0.9449	0.9328	0.9434	0.9371
KNN	0.5003	0.4559	0.6999	0.6551

Table 2: Models F1-Score

4.1.2 F1-Score

F1-score was used as a complementary metric to account for imbalances between precision and recall. Here, too, ANN led the baseline evaluation with a score of 0.9449, confirming its ability to consistently detect posture classes. When BMI was introduced, XGB again achieved the best performance ever, reaching a score of 0.9750 with filtered data, closely followed by RF and ANN, as shown in Table 2. The addition of BMI helped these models become more sensitive to postures influenced by weight distribution. In contrast, SVM's F1-score dropped significantly across all BMI-inclusive configurations, likely due to limitations in handling heterogeneous feature types without prior normalization.

4.2 Internal Variables: Center of Pressure and Percentage of Pressure

Including the CoP and Percentage of Pressure as spatial features improved classification performance by capturing balance and contact area information. The RF achieved the best accuracy and F1-score (0.98) using raw data with Percentage of Pressure, confirming its robustness. KNN and ANN also performed well with CoP-based inputs, while SVM proved highly sensitive to noise. Overall, Percentage of Pressure enhanced model performance, particularly for ensemble-based methods.

5 Discussion

The results show that combining pressure maps with derived features significantly improves posture classification. Tree-based models (RF, XGB) achieved the best overall performance, especially when BMI and spatial features (CoP, Percentage of Pressure) were added to raw data. ANN also showed strong adaptability, particularly with filtered CoP inputs, while SVM proved sensitive to noisy, unstructured data. Overall, ensemble methods integrated heterogeneous features more effectively, whereas KNN was more affected by noise and filtering.

6 Conclusion

This work presented a machine learning framework for automatic classification of sitting posture using pressure sensor data. Through extensive testing across multiple algorithms and feature configurations, we demonstrated that enriching raw image data with biometric and spatial features, can significantly improve classification performance.

The best results were achieved using XGB and ANN models with minimal filtering, confirming their suitability for real-time ergonomic applications. Future work will explore real-time deployment, temporal posture tracking, and the inclusion of dynamic features for broader ergonomic monitoring.

References

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